

YINGJIN LI

M.Phil. Student, The Hong Kong University of Science and Technology (Guangzhou)

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Research Interests: Visual Anomaly Detection · Open-World Visual Understanding
Multimodal Learning · Vision-Language Models

Education

The Hong Kong University of Science and Technology (Guangzhou)

M.Phil., Data Science and Analytics

Sep 2025 - Jun 2027 (expected)

Thesis Supervisor: Prof. Lei Zhu

Recipient of the Red Bird Scholarship (RMB 240,000).

Wuhan Textile University

B.Eng., Data Science and Big Data Technology

Sep 2021 - Jun 2025

Average score: 86.7/100 · Academic ranking: Top 5% · Overall ranking: 1st in major

Publication List

1. ROAD: Rule-Grounded Context-Aware Open-World Driver Anomaly Detection

Yingjin Li, Shuaibo Li, Wei Ma, Zhijie Qiu, Fuxiang Zhai, Sixiang Chen, Zhaolu Kang, and Lei Zhu.

First Author · *Advances in Neural Information Processing Systems (NeurIPS)*, 2026 · Accepted

2. Fashion Image Retrieval Based on Multimodal Features Enhancement and Fusion

Yingjin Li, Shufan He, Zhaojing Wang, Jin Huang, Xinrong Hu, and Li Li.

First Author · *Computer Animation and Social Agents (CASA 2024)* · Published · Springer, CCIS 2374, pp. 118-131, 2025.

3. PriSM-AD: Learning Common Normal Priors and Anomaly Semantics for Diffusion-Based Anomaly Detection

Yingjin Li, Sixiang Chen, Wenxue Li, Fugee Tsung, and Lei Zhu.

First Author · *IEEE Transactions on Multimedia (TMM)* · Under Review

Research Experience

ROAD · Multimodal Driver Anomaly Detection [1]

Apr 2026 - Present

- Designed and implemented a **multimodal anomaly detection framework** that integrates driver video, driving context, and explicit behavioral rules to identify previously unseen driver anomalies.
- Developed **rule-grounded reasoning** that translates domain knowledge into structured guidance for multimodal large language models, supporting explanations of why a behavior is abnormal beyond predefined action categories.
- Integrated **context-aware visual reasoning** to distinguish visually similar behaviors across driving situations and account for context-dependent safety rules in open-world anomaly understanding.
- Built an evaluation pipeline across driver-monitoring datasets with free-form anomaly descriptions, human assessments, and MLLM baselines to assess open-world generalization and model-human agreement.

PriSM-AD · Diffusion-Based Anomaly Detection [3]

- Designed a **two-stage diffusion framework** that separates learning common normal visual priors from anomaly localization, combining normality modeling with semantic cues for industrial defect detection.
- Developed **prior-guided reconstruction** to recover normal appearance while suppressing anomalous structures, providing a normal reference for identifying subtle defects.
- Integrated **anomaly semantics and reconstruction cues** to support both image-level anomaly detection and pixel-level localization across industrial categories.
- Analyzed reconstruction quality, cross-category generalization, anomaly localization, and inference efficiency.

Honors & Awards

- National Scholarship**, Ministry of Education of China, 2024
- First Prize, Hubei Province**, China Mathematical Contest in Modeling
- National First Prize**, Teddy Cup Data Mining Challenge
- National First Prize**, Huazhong Cup Mathematical Modeling Challenge